

Appendix M

Miscellaneous examples

M.1 Gibbs ensemble for dense liquids

Example 32 (Dense liquids). At high densities, the number of exchange steps can become very large and the simulation requires a significant amount of CPU time. This problem occurs also in conventional grand-canonical Monte Carlo simulations. Various methods, which are used to extend simulations in the grand-canonical ensemble to higher densities, can also be used in the Gibbs ensemble. An example of such a technique is the so-called excluded volume map sampling. This technique, based on the ideas of Deitrick et al. [782] and Mezei [188], has been applied to the Gibbs ensemble by Stapleton and Panagiotopoulos [783]. Before calculating the energy of the particle that has to be inserted, a map is made of the receiving subsystem, by dividing this subsystem into small boxes that can contain at most one particle. Each box carries a label that indicates whether it is empty or contains a particle. This map can then be used as a lookup table to check whether there is “space” for the particle to be inserted. If such a space is not available, the trial configuration can be rejected immediately. When using the excluded-volume map, some additional bookkeeping is needed to guarantee detailed balance (see [188] for further details).

M.2 Free energy of a nitrogen crystal

Example 33 (Free energy of a nitrogen crystal). For an orientationally disordered solid, we can use an approach similar to that used for atomic crystals. Let us consider an orientationally disordered molecular solid. We transform this solid into a state of known free energy in two stages [404,407]. First we couple the molecules in the solid with harmonic springs to their lattice sites. But in contrast to the method described earlier, we leave the original intramolecular interactions unaffected. Subsequently, we expand this “interacting Einstein crystal” to zero density (see Fig. M.1). Due to the coupling to the lattice, the crystal cannot melt on expansion but keeps its original structure. In the low-density limit, all intermolecular interactions vanish and the system behaves as an ideal Einstein crystal. This scheme of calculating an absolute free energy is referred to as the *lattice-coupling-expansion method* [404,407].

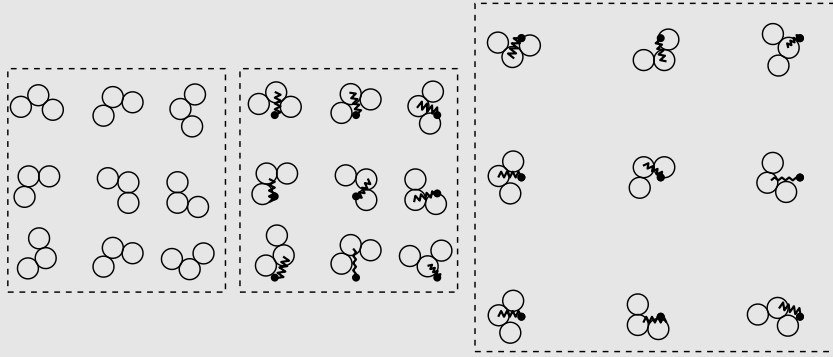


FIGURE M.1 Schematic drawing of the lattice-coupling-expansion method for calculating the free energy of a molecular solid: the first step is the coupling to an Einstein crystal (denoted by the black dots) and the second step is the expansion to zero density.

During the first stage of the thermodynamic integration, the potential energy function $\tilde{\mathcal{U}}_1$ contains both the original intermolecular potential and the harmonic coupling to the lattice:

$$\tilde{\mathcal{U}}_1(\mathbf{r}^N, \mathbf{\Omega}^N; \lambda) = \mathcal{U}(\mathbf{r}^N, \mathbf{\Omega}^N) + \lambda \sum_{i=1}^N \alpha (\mathbf{r}_i - \mathbf{r}_{0,i})^2, \quad (\text{M.2.1})$$

where $\mathbf{\Omega}_i$ denotes the orientation of particle i , $\mathbf{r}_i$ its center-of-mass position, and $\mathbf{r}_{0,i}$ the lattice site of particle i . For convenience, we have assumed that all lattice sites are equivalent. We therefore use the same value of the coupling constant α for all sites. In most molecular solids, several nonequivalent molecules may be in a unit cell. In that case different coupling constants may be chosen for all distinct lattice sites.

The change in free energy associated with switching on the harmonic springs is given by Eq. (8.4.8):

$$\Delta F_1 = F(\lambda = 1) - F_{\text{mol sol}} = \int_0^1 d\lambda \left\langle \sum_{i=1}^N \alpha (\mathbf{r}_i - \mathbf{r}_{0,i})^2 \right\rangle_{\lambda}. \quad (\text{M.2.2})$$

It is reasonable to expect that the integrand in Eq. (M.2.2) is a smooth function of λ , as the mean-squared displacement decreases monotonically with increasing λ .

During the second stage of the thermodynamic integration, all molecules remain harmonically coupled to their (Einstein) lattice sites, but this reference lattice is expanded uniformly to zero density. In what follows, we assume for convenience that the intermolecular potential is pairwise additive:

$$\mathcal{U}(\mathbf{r}^N, \mathbf{\Omega}^N) = \sum_{i < j}^N \mathcal{U}_{\text{pair}}(r_{ij}, \mathbf{\Omega}_i, \mathbf{\Omega}_j),$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the distance between the centers of mass of the molecules i and j . When we expand the system uniformly with a factor γ , the coordinates of lattice sites are given by $\gamma \mathbf{r}_{0,i}$. When the lattice is expanded, the intermolecular interactions between the molecules change. This is best seen by expressing the center-of-mass coordinate of particle i as

$$\mathbf{r}_i = \gamma \mathbf{r}_{0,i} + \Delta \mathbf{r}_i,$$

where $\Delta \mathbf{r}_i$ is the position with respect to a lattice site of the expanded Einstein crystal. For the distance between two particles, we can write

$$r_{ij} = |\gamma(\mathbf{r}_{0,i} - \mathbf{r}_{0,j}) + \Delta \mathbf{r}_i - \Delta \mathbf{r}_j|. \quad (\text{M.2.3})$$

In terms of these coordinates, the potential energy of the expanded crystal is

$$\tilde{\mathcal{U}}_{\text{II}}(\mathbf{r}^N, \boldsymbol{\Omega}^N; \gamma) = \sum_{i < j}^N \mathcal{U}_{\text{pair}}(r_{ij}, \boldsymbol{\Omega}_i, \boldsymbol{\Omega}_j) + \sum_{i=1}^N \alpha(\Delta \mathbf{r}_i)^2. \quad (\text{M.2.4})$$

Note that only the *intermolecular* interactions depend on γ (through r_{ij}). The derivative of the free energy with respect to γ is

$$\left\langle \frac{\partial \tilde{\mathcal{U}}}{\partial \gamma} \right\rangle = \left\langle \sum_{i < j}^N \frac{\partial \mathcal{U}_{\text{pair}}(r_{ij}, \boldsymbol{\Omega}_i, \boldsymbol{\Omega}_j)}{\partial \gamma} \right\rangle = \left\langle \sum_{i < j}^N \frac{\partial \mathcal{U}_{\text{pair}}(r_{ij}, \boldsymbol{\Omega}_i, \boldsymbol{\Omega}_j)}{\partial r_{ij}} \frac{\partial r_{ij}}{\partial \gamma} \right\rangle.$$

Only at this stage do we make use of the assumption that the intermolecular potential is pairwise additive. We stress, however, that the assumption is not essential—it just yields a simpler form for the final expression.

From Eq. (M.2.3) it follows that

$$\frac{\partial r_{ij}}{\partial \gamma} = \frac{\mathbf{r}_{ij} \cdot \mathbf{r}_{ij}^0}{r_{ij}}.$$

The change in free energy due to the uniform expansion is

$$\begin{aligned} \Delta F_{\text{II}} &= F_{\text{Ein}}^{\text{mol}} - F(\lambda = 1, \gamma = 1) \\ &= \int_1^\infty d\gamma \left\langle \frac{\partial \tilde{\mathcal{U}}}{\partial \gamma} \right\rangle \\ &= \int_1^\infty d\gamma \left\langle \sum_{i < j}^N \frac{\partial \mathcal{U}_{\text{pair}}(r_{ij}, \boldsymbol{\Omega}_i, \boldsymbol{\Omega}_j)}{\partial r_{ij}} \frac{\mathbf{r}_{ij} \cdot \mathbf{r}_{ij}^0}{r_{ij}} \right\rangle. \end{aligned} \quad (\text{M.2.5})$$

Note that if, in the preceding equation, we replace $\mathbf{r}_{ij}^0$ by r_{ij} , the expression in angular brackets reduces to the virial. We therefore refer to the integrand in Eq. (M.2.5) as a modified virial.

Finally, we must evaluate the free energy of the reference state: an Einstein crystal consisting of noninteracting molecules. As the intramolecular contribution to the free energy is a constant that depends only on temperature, we

shall ignore it. The expression for the total free energy of the molecular solid then becomes

$$F_{\text{mol sol}} = F_{\text{Ein}}^{\text{mol}} - \Delta F_{\text{I}} - \Delta F_{\text{II}}. \quad (\text{M.2.6})$$

For molecular solids with (partial) orientational order, a similar scheme can be used. We transform the solid by imposing a coupling of the centers of mass of the molecules together with a coupling of the *ordered* degrees of freedom. With the combined coupling the solid is expanded. The details of this scheme depend on the nature of the orientational order.

M.3 Zeolite structure solution

Example 34 (Zeolite structure solution). Falcioni and Deem [550] used parallel tempering as a method to find zeolite structures. Zeolites are inorganic crystals with a known chemical structure. The interesting aspect of these materials is that there are about 118 different known zeolite structures. But many more zeolite structures are theoretically possible. If one makes a new zeolite, one often has a powder and it is much more time consuming, if possible at all, to make a sufficiently large crystal from which the crystal structure can be determined.

The problem that Falcioni and Deem were interested in was finding the zeolite structure that corresponds best to an experimental powder diffraction pattern. Experimentally, it is well known that in zeolites the angles between three successive T-sites (positions of the Si atoms in an all-silica zeolite) are clustered around 109.5° . Falcioni and Deem assumed that any new zeolite structure should have angles with a similar distribution. In a similar way, rules were deduced for the average bond distance and other properties of the structures. These rules were translated into a pseudo-Hamiltonian; this Hamiltonian has a high value for those crystal structures that do not resemble the characteristics of a zeolite and a low value for those crystals that have zeolite-like characteristics. An additional term was added to this structural pseudo-Hamiltonian that indicated the difference between the experimental and the calculated X-ray powder pattern of the new crystal structure. Falcioni and Deem then assumed that the unknown zeolite structure corresponds to the structure that minimizes this combined Hamiltonian. Various Monte Carlo rules were invented to change the crystal structure, and a new structure was accepted or rejected by using this pseudo-Hamiltonian. Since this pseudo-Hamiltonian has many local minima, parallel tempering was used to find the optimal structure.

To test their method, Falcioni and Deem used the information of 32 zeolite structures to generate the pseudo-Hamiltonian. This information was sufficient to solve the structure of the remainder of all known zeolite structures.